SMM313 Numerical Methods: Applications

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Exercise 1. Asian Options

We are pricing a standard arithmetic Asian call (A_n), its lower bound (LB_n) and the geometric Asian call (G_n), under the geometric Brownian motion model. The asset price at the k-th of n equally spaced monitoring dates is denoted S_k, the continuously compounded risk-free rate r, the strike is K and T time to maturity. The discounted payoff functions are:

$$A_n = e^{-rT} \max\left(\frac{1}{n}\sum_{k=1}^{n} S_k - K,\ 0\right), \tag{1}$$

$$\mathrm{LB}_n = e^{-rT}\left(\frac{1}{n}\sum_{k=1}^{n} S_k - K\right)\mathbf{1}_{\left\{\left(\prod_{k=1}^{n} S_k\right)^{1/n} > K\right\}}, \tag{2}$$

$$G_n = e^{-rT} \max\left(\left(\prod_{k=1}^{n} S_k\right)^{1/n} - K,\ 0\right). \tag{5}$$

Question 1.1. Sensitivity Estimators

Part (i): LR first-order sensitivity estimators w.r.t. S_0

Since all three payoffs are path-dependent, the relevant delta score is

$$\frac{d\ln g_{S_0}(S_1,\ldots,S_n)}{dS_0} = \frac{\zeta(S_1 \mid S_0)}{S_0\,\sigma\sqrt{\delta}}, \qquad \zeta(S_1 \mid S_0) = \frac{\ln(S_1/S_0) - (r-\sigma^2/2)\,\delta}{\sigma\sqrt{\delta}},$$

where $\delta = T/n$. Hence, for any discounted payoff $C = f(S_1,\ldots,S_n)$, the LR delta representation is $\mathbb{E}\big[f(S_1,\ldots,S_n)\cdot\zeta(S_1 \mid S_0)/(S_0\sigma\sqrt{\delta})\big]$, giving:

(a) Arithmetic Asian call option with discounted payoff (1):

$$\frac{d\,\mathbb{E}(A_n)}{dS_0} = \mathbb{E}\left[e^{-rT}\max\left(\frac{1}{n}\sum_{k=1}^{n} S_k - K,\ 0\right)\frac{\zeta(S_1 \mid S_0)}{S_0\,\sigma\sqrt{\delta}}\right].$$

(b) Lower bound with discounted payoff (2):

$$\frac{d\,\mathbb{E}(\mathrm{LB}_n)}{dS_0} = \mathbb{E}\left[e^{-rT}\left(\frac{1}{n}\sum_{k=1}^{n} S_k - K\right)\mathbf{1}_{\left\{\left(\prod_{k=1}^{n} S_k\right)^{1/n} > K\right\}}\frac{\zeta(S_1 \mid S_0)}{S_0\,\sigma\sqrt{\delta}}\right].$$

(c) Geometric Asian call option with discounted payoff (5):

$$\frac{d\,\mathbb{E}(G_n)}{dS_0} = \mathbb{E}\left[e^{-rT} \max\left(\left(\prod_{k=1}^{n} S_k \right)^{1/n} - K,\ 0 \right) \frac{\zeta(S_1 \mid S_0)}{S_0\, \sigma \sqrt{\delta}} \right].$$

Part (ii): Existence of a PW estimator for LB_n

The pathwise method requires the payoff to be Lipschitz continuous in S_0 so that differentiation and expectation can be interchanged. The indicator in (2) introduces a jump discontinuity at the boundary where the geometric mean equals K, as illustrated in Figure 1. Since the payoff is not continuous at this boundary, a PW first-order sensitivity estimator for LB_n does **not** exist.

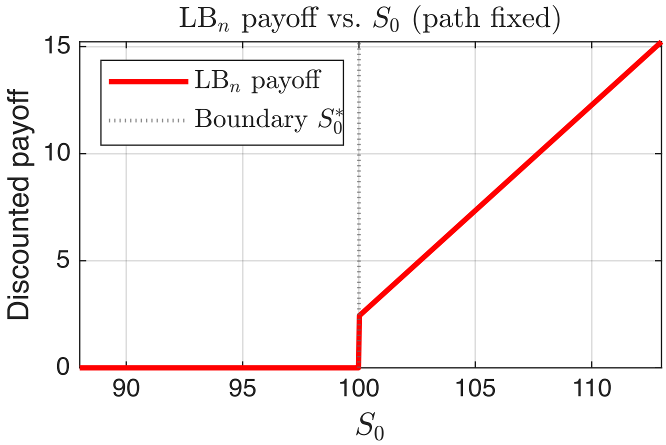

Figure 1: LB_n payoff as a function of S_0 for a fixed simulated path. The jump discontinuity at S_0^* (where the geometric mean equals K) confirms that no pathwise estimator exists.

Question 1.2. Exact Sensitivity Formulae

We differentiate the Curran (1994) formula for $E(\mathrm{LB}_n)$ and the Kemna–Vorst (1990) formula for $E(G_n)$ with respect to S_0. Throughout, $N(\cdot)$ and $\phi(\cdot)$ denote the standard normal CDF and PDF. The closed-form prices are

$$E(\mathrm{LB}_n) = \frac{S_0 e^{-rT}}{n} \sum_{k=1}^{n} e^{\mu_k + \sigma_k^2/2} N(b + a_k) - K e^{-rT} N(b), \tag{3}$$

$$E(G_n) = S_0\, \alpha\, N(d) - K e^{-rT} N\left(d - \bar{\sigma}\sqrt{\bar{T}}\right), \tag{6}$$

where

$$\alpha = e^{(r - \sigma^2/2 + \bar{\sigma}^2/2)\bar{T} - rT},$$

and

$$\bar{\sigma} = \sigma \sqrt{\frac{2n+1}{3n}}, \qquad \bar{T} = \frac{(n+1)\delta}{2}, \qquad b = \frac{\ln(S_0/K) + (r - \sigma^2/2)\bar{T}}{\bar{\sigma}\sqrt{\bar{T}}}, \qquad d = b + \bar{\sigma}\sqrt{\bar{T}}. \tag{4}$$

Also, for $k = 1, \ldots, n$,

$$\mu_k = \left(r - \frac{\sigma^2}{2}\right) k\delta, \qquad \sigma_k = \sigma\sqrt{k\delta}, \qquad a_k = \frac{\sigma\sqrt{\delta}\,\sqrt{k\big(n+1-(k+1)/2\big)}}{\sqrt{n(n+1)(2n+1)/6}}.$$

Since μ_k, σ_k, a_k, $\bar{\sigma}$ and $\bar{T}$ do not depend on S_0, only b and d vary with S_0. Hence

$$\frac{db}{dS_0} = \frac{dd}{dS_0} = \frac{1}{S_0\,\bar{\sigma}\sqrt{\bar{T}}}.$$

Sensitivity of $E(\mathrm{LB}_n)$

Differentiating (3) term by term,

$$\frac{d}{dS_0}E(\mathrm{LB}_n) = \frac{d}{dS_0}\left[\frac{S_0 e^{-rT}}{n}\sum_{k=1}^{n} e^{\mu_k+\sigma_k^2/2}N(b+a_k)\right] - \frac{d}{dS_0}\left[Ke^{-rT}N(b)\right].$$

Using the product rule on the first term and the chain rule on $N(b+a_k)$,

$$\frac{d}{dS_0}\left[\frac{S_0 e^{-rT}}{n}e^{\mu_k+\sigma_k^2/2}N(b+a_k)\right] = \frac{e^{-rT}}{n}e^{\mu_k+\sigma_k^2/2}N(b+a_k) + \frac{S_0 e^{-rT}}{n}e^{\mu_k+\sigma_k^2/2}\phi(b+a_k)\frac{db}{dS_0}.$$

Also, $\frac{d}{dS_0}\left[Ke^{-rT}N(b)\right] = Ke^{-rT}\phi(b)\,\frac{db}{dS_0}$. Substituting $\frac{db}{dS_0} = \frac{1}{S_0\bar{\sigma}\sqrt{\bar{T}}}$ and summing over k:

$$\boxed{\frac{dE(\mathrm{LB}_n)}{dS_0} = \frac{e^{-rT}}{n}\sum_{k=1}^{n} e^{\mu_k+\sigma_k^2/2}N(b+a_k) + \frac{e^{-rT}}{S_0\,\bar{\sigma}\sqrt{\bar{T}}}\left[\frac{S_0}{n}\sum_{k=1}^{n} e^{\mu_k+\sigma_k^2/2}\phi(b+a_k) - K\phi(b)\right].}$$

Sensitivity of $E(G_n)$

Differentiating (6),

$$\frac{dE(G_n)}{dS_0} = \alpha\,N(d) + S_0\,\alpha\,\phi(d)\,\frac{dd}{dS_0} - Ke^{-rT}\phi\Big(d-\bar{\sigma}\sqrt{\bar{T}}\Big)\,\frac{dd}{dS_0}.$$

Since $\frac{dd}{dS_0} = \frac{1}{S_0\bar{\sigma}\sqrt{\bar{T}}}$, the S_0 factors cancel:

$$\boxed{\frac{dE(G_n)}{dS_0} = \alpha\,N(d) + \frac{\alpha\,\phi(d) - \frac{K}{S_0}e^{-rT}\phi\Big(d-\bar{\sigma}\sqrt{\bar{T}}\Big)}{\bar{\sigma}\sqrt{\bar{T}}}.}$$

Question 1.3. Exact Numerical Results

Using formulae (3) and (6) and the exact sensitivity expressions derived in Question 1.2, we compute $E(\mathrm{LB}_n)$ and $E(G_n)$ together with their first-order sensitivities with respect to S_0, for all nine (K, n) pairs. The parameter values are $S_0 = 100$, $r = 0.04$, $\sigma = 0.3$, $T = 1$. Results are reported in Tables 1 and 2. (See `question3.m` for the implementation).

K	n	$E(\mathrm{LB}_n)$	$dE(\mathrm{LB}_n)/dS_0$
90	4	15.032485	0.755865
90	12	14.135865	0.765755
90	50	13.798093	0.770449
100	4	9.244950	0.576484
100	12	8.234544	0.567149
100	50	7.848960	0.563349
110	4	5.290978	0.396819
110	12	4.370197	0.368745
110	50	4.026784	0.356836

Table 1: Expected lower bound $E(\mathrm{LB}_n)$ and its sensitivity $dE(\mathrm{LB}_n)/dS_0$ from the Curran formula (3) and the derivative derived in Question 1.2.

K	n	$E(G_n)$	$dE(G_n)/dS_0$
90	4	14.524887	0.742478
90	12	13.602387	0.751305
90	50	13.263128	0.755796
100	4	8.831478	0.562878
100	12	7.802060	0.552774
100	50	7.415534	0.548990
110	4	4.971044	0.383556
110	12	4.038233	0.354391
110	50	3.694512	0.342256

Table 2: Geometric Asian call price $E(G_n)$ and its sensitivity $dE(G_n)/dS_0$ from the Kemna–Vorst formula (6) and the derivative derived in Question 1.2.

Across both tables, $E(\mathrm{LB}_n)$ and $E(G_n)$ decrease as K increases, reflecting a move further out-of-the-money, and for the parameter values considered here both quantities also decrease as n increases. For all (K, n), $E(\mathrm{LB}_n) > E(G_n)$, consistent with the AM–GM inequality applied path-wise. The gap is small (typically 3–6%), confirming that the geometric average is a tight proxy for the arithmetic average.

The sensitivity $dE(\mathrm{LB}_n)/dS_0$ decreases as K rises, reflecting the declining probability of exercise. Its behaviour with n depends on moneyness: for deep in-the-money options ($K = 90$) the delta increases slightly with n, as finer averaging concentrates the distribution of $\bar{S}$ above the strike; for at-the-money and out-of-the-money options ($K = 100, 110$) the delta decreases with n, as averaging reduces the variance of $\bar{S}$ and flattens the payoff profile.

Question 1.4. Monte Carlo Simulation with Control Variates

We use $M = 10^5$ simulated GBM paths to estimate both the arithmetic Asian option price $E(A_n)$ and its LR first-order sensitivity $dE(A_n)/dS_0$, for all nine (K, n) pairs. Two control variates are applied in each case:

(a) LB_n for prices, and its LR delta for sensitivities, with closed-form expectations $E(\mathrm{LB}_n)$ and $dE(\mathrm{LB}_n)/dS_0$ from Questions 1.2 and 1.3;

(b) G_n for prices, and its LR delta for sensitivities, with closed-form expectations $E(G_n)$ and $dE(G_n)/dS_0$ from Questions 1.2 and 1.3.

For any target Y and control X with known expectation $E[X]$, the control-variate estimator and optimal coefficient are

$$\hat{Y}^{\mathrm{cv}} = \bar{Y} - \hat{\beta}\,(\bar{X} - E[X]), \qquad \hat{\beta} = \frac{\widehat{\mathrm{Cov}}(Y,\,X)}{\widehat{\mathrm{Var}}(X)}.$$

The efficiency ratio comparing method (a) to method (b) is

$$\mathcal{E}(K,n) = \frac{\hat{t}_a\,\hat{\sigma}_a^2}{\hat{t}_b\,\hat{\sigma}_b^2},$$

where $\hat{\sigma}^2$ is the squared standard error and $\hat{t}$ the measured runtime; $\mathcal{E} < 1$ implies method (a) is more efficient than method (b). Because both approaches use the same simulated GBM paths, the reported runtimes measure the computational cost of the corresponding control-variate estimation step. The efficiency ratios are reported in the last column of each table. (See `question4.m`)

Parts (i)–(ii): Price and sensitivity results

Both methods estimate the same target ($E(A_n)$ for prices, $dE(A_n)/dS_0$ for deltas); they differ only in variance. Table 3 reports price estimates and Table 4 reports LR delta estimates, with the efficiency ratio in the final column of each.

		Method (a): control LB_n				Method (b): control G_n				
K	n	Price	SE	CI_L	CI_U	Price	SE	CI_L	CI_U	$\mathcal{E}$
90	4	15.0370	0.000209	15.0366	15.0374	15.0376	0.002042	15.0336	15.0416	0.0105
90	12	14.1401	0.000176	14.1397	14.1404	14.1393	0.001812	14.1358	14.1429	0.0095
90	50	13.8023	0.000175	13.8020	13.8027	13.8039	0.001730	13.8006	13.8073	0.0102
100	4	9.2499	0.000224	9.2495	9.2503	9.2516	0.001927	9.2478	9.2554	0.0135
100	12	8.2382	0.000166	8.2379	8.2386	8.2380	0.001675	8.2347	8.2413	0.0098
100	50	7.8523	0.000152	7.8520	7.8526	7.8526	0.001568	7.8495	7.8557	0.0094
110	4	5.2961	0.000242	5.2956	5.2966	5.2979	0.001815	5.2943	5.3015	0.0177
110	12	4.3751	0.000211	4.3747	4.3755	4.3754	0.001577	4.3723	4.3785	0.0178
110	50	4.0318	0.000214	4.0314	4.0322	4.0321	0.001461	4.0292	4.0350	0.0214

Table 3: Arithmetic Asian call option price estimates $E(A_n)$ using control variates (a) LB_n and (b) G_n, with standard errors, 95% confidence intervals, and efficiency ratio $\mathcal{E}(K,n)$. Both methods estimate the same target; method (a) achieves substantially lower standard errors throughout.

		Method (a): control LR(LB_n)				Method (b): control LR(G_n)				
K	n	Delta	SE	CI_L	CI_U	Delta	SE	CI_L	CI_U	$\mathcal{E}$
90	4	0.755903	0.000017	0.755870	0.755936	0.755942	0.000157	0.755634	0.756250	0.0114
90	12	0.765872	0.000025	0.765823	0.765921	0.765686	0.000232	0.765232	0.766141	0.0116
90	50	0.770609	0.000042	0.770527	0.770691	0.771348	0.000433	0.770499	0.772198	0.0092
100	4	0.576555	0.000023	0.576510	0.576600	0.576525	0.000147	0.576236	0.576814	0.0249
100	12	0.567170	0.000026	0.567119	0.567221	0.567005	0.000216	0.566581	0.567428	0.0142
100	50	0.563406	0.000040	0.563328	0.563484	0.564046	0.000398	0.563265	0.564827	0.0101
110	4	0.396798	0.000019	0.396761	0.396835	0.396805	0.000133	0.396545	0.397065	0.0213
110	12	0.368643	0.000027	0.368590	0.368696	0.368581	0.000199	0.368191	0.368971	0.0181
110	50	0.356663	0.000052	0.356561	0.356765	0.357096	0.000372	0.356367	0.357825	0.0194

Table 4: Arithmetic Asian call LR delta estimates $dE(A_n)/dS_0$ using control variates (a) LR delta of LB_n and (b) LR delta of G_n, with standard errors, 95% confidence intervals, and efficiency ratio $\mathcal{E}(K, n)$.

Part (iii): Commentary

Accuracy of estimates. Accuracy is assessed via the 95% CI half-width $h = (CI_U - CI_L)/2$, with the approximate number of correct decimal digits given by $-\log_{10}(h)$. Under method (a), price half-widths are of order 4×10^{-4}, giving approximately 3.4 correct decimal digits. By method (b), half-widths are of order 4×10^{-3}, giving approximately 2.4 digits. Method (a) achieves about one additional decimal digit of accuracy for the same simulation budget. For sensitivities, method (a)'s delta half-widths are of order 10^{-5} versus 10^{-4} under method (b), the same one-digit improvement.

Effect of the two control variates. Method (a) produces standard errors roughly ten times smaller than method (b) for both prices and sensitivities. For prices, this reflects the very high positive correlation between LB_n and A_n: both depend on the same arithmetic path average, so LB_n absorbs almost all simulation variance with $\hat{\beta} \approx 1$. The geometric payoff G_n is a weaker proxy because the arithmetic and geometric averages diverge as n grows.

For sensitivities, both methods must use the LR estimator for the target $dE(A_n)/dS_0$, since no closed-form pathwise derivative of the arithmetic Asian payoff exists in general. For the control variate, method (a) uses the LR delta of LB_n, which is forced: LB_n contains the indicator $\mathbf{1}_{\{G_n^{1/n} > K\}}$, making it discontinuous in S_0 and ruling out a pathwise estimator (see Question 1.1(ii)). Method (b) uses the LR delta of G_n, where a pathwise alternative exists as $G_n = e^{-rT} \max(G_n^{1/n} - K, 0)$ is continuous and differentiable in S_0 almost surely. Thus, the PW estimator $e^{-rT}(G_n^{1/n}/S_0)\mathbf{1}_{\{G_n^{1/n} > K\}}$ is valid. However, the LR delta of G_n uses the same score as method (a) and shares the same simulation paths, making it a natural and consistent choice. The LR delta of LB_n dominates because both LR delta samples multiply the same score function $\zeta(S_1|S_0)/(S_0\sigma\sqrt{\delta})$ by highly correlated payoffs, giving very high sample correlation between the control and the target.

Efficiency gains: prices vs. sensitivities. All $\mathcal{E}(K, n) < 1$, confirming method (a) dominates throughout. For prices, $\mathcal{E}$ ranges from roughly 0.009 to 0.021, implying between 48 and 110 times more efficient. For sensitivities, $\mathcal{E}$ ranges from roughly 0.009 to 0.025, a similar level. The efficiency gains are therefore comparable between prices and deltas for a given (K, n).

Effect of K and n. For fixed n, $\mathcal{E}(K, n)$ tends to increase slightly as K rises, reflecting a mild weakening of the correlation between A_n and LB_n as the option moves further out-of-the-money.

For fixed K, the effect of varying n is modest and non-monotone under both methods.

Proximity of $E(\mathrm{LB}_n)$ to the arithmetic price. From Table 3, the estimates from method (a) and method (b) agree to three or more decimal places in every case, confirming both are correctly estimating $E(A_n)$. Comparing these estimates to the exact $E(\mathrm{LB}_n)$ values in Table 1, the two are extremely close: for example at $K = 100$, $n = 12$, $E(\mathrm{LB}_{12}) = 8.2345$ versus the MC estimate of 8.2382, a relative error of only 0.05%. The gap widens slightly as K increases and as n increases, but remains small throughout, confirming that LB_n is a tight lower bound and an effective control variate.

Question 1.5. Convergence of $E(\mathrm{LB}_n)$ to the Continuous Limit

Thompson (1999) derives the closed-form continuous-monitoring limit

$$E(\mathrm{LB}_\infty) = \frac{S_0}{T}\int_0^T e^{-r(T-t)} N\left(\frac{-\gamma + \sigma t - \sigma t^2/(2T)}{\sqrt{T/3}}\right) dt \; - \; K e^{-rT} N\left(\frac{-\gamma}{\sqrt{T/3}}\right), \tag{7}$$

where $\gamma = \big(\ln(K/S_0) - (r - \sigma^2/2)T/2\big)/\sigma$. Formula (7) is evaluated numerically using MATLAB's `integral()` function, giving $E(\mathrm{LB}_\infty) = 7.72696136$ for the parameters $S_0 = 100$, $r = 0.04$, $\sigma = 0.3$, $K = 100$, $T = 1$.

Table 5 reports $E(\mathrm{LB}_n)$ computed from formula (3) for $n = 2^2, 2^3, \ldots, 2^{10}$, alongside the absolute error and the ratio of successive errors. (See `question5.m`.)

n	$E(\mathrm{LB}_n)$	$\lvert E(\mathrm{LB}_n) - E(\mathrm{LB}_\infty)\rvert$	Ratio
4	9.24494951	1.51798816	—
8	8.48767043	0.76070908	1.995
16	8.10782467	0.38086331	1.997
32	7.91753218	0.19057082	1.999
64	7.82228320	0.09532184	1.999
128	7.77463160	0.04767024	2.000
256	7.75079884	0.02383748	2.000
512	7.73888069	0.01191933	2.000
1024	7.73292117	0.00595982	2.000

Table 5: Convergence of $E(\mathrm{LB}_n)$ to $E(\mathrm{LB}_\infty) = 7.72696136$. The ratio column reports $\lvert E(\mathrm{LB}_{n/2}) - E(\mathrm{LB}_\infty)\rvert \,/\, \lvert E(\mathrm{LB}_n) - E(\mathrm{LB}_\infty)\rvert$. The Ratio column is undefined for n=4 as there is no preceding entry n=2

The sequence decreases monotonically from above, converging towards $E(\mathrm{LB}_\infty)$ as the discrete grid becomes finer. The ratio of successive absolute errors converges rapidly to exactly 2, confirming an $O(1/n)$ convergence rate: each doubling of n halves the error precisely. This is consistent with a Riemann-sum interpretation of formula (3): as $\delta = T/n \to 0$, the discrete sum approximates the continuous integral in (7) with first-order accuracy in δ. This rate is notably faster than the $O(n^{-1/2})$ rate observed for the discretely monitored barrier option in Exercise 2, where the error arises from missed Brownian bridge crossings rather than quadrature approximation.

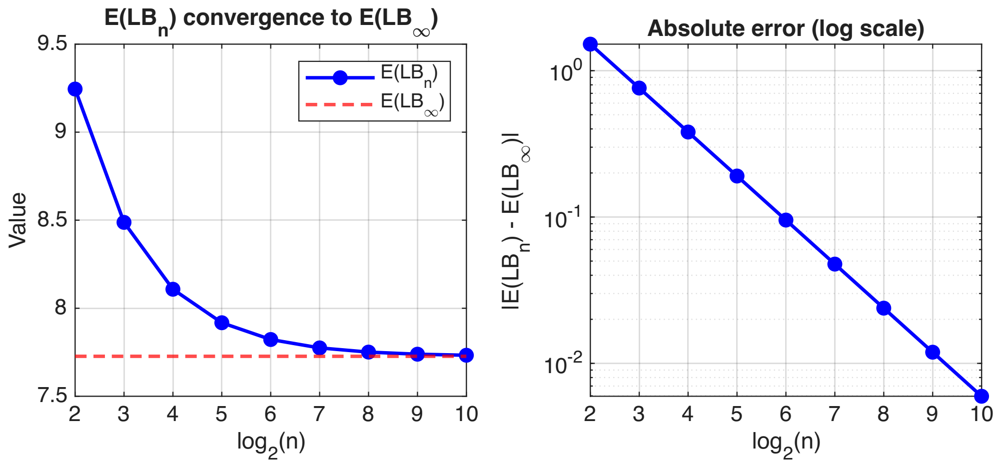

Figure 2: Left: $E(\mathrm{LB}_n)$ (blue) decreasing monotonically to the continuous benchmark $E(\mathrm{LB}_\infty) = 7.72696$ (red dashed) as n increases. Right: absolute error on a log scale, with a reference line of slope -1 confirming $O(1/n)$ convergence.

Exercise 2. Discretely Monitored Up-and-Out Barrier Call Option

We consider a discretely monitored up-and-out barrier call (UOC) option on an asset S following geometric Brownian motion. The option is monitored at n equally spaced dates separated by $\delta = T/n$. If the asset reaches or exceeds the upper barrier U at any monitoring date the option is knocked out; otherwise it pays the standard call payoff at maturity. The discounted payoff is

$$C_n = e^{-rT} \max(S_n - K,\, 0)\, \mathbf{1}_{\{\max(S_0, S_1, \ldots, S_n) < U\}}.$$

The price $E(C_n)$ has no closed form and is estimated by Monte Carlo simulation with antithetic variates. The continuously monitored limit ($n \to \infty$) admits the closed-form expression from Shreve (2004):

$$f_{\mathrm{UOC}}(T, S_0, K, U, \sigma, r) = \mathcal{A} - \mathcal{B}, \tag{8}$$

where

$$\begin{aligned}
\mathcal{A} &= S_0\big[N(d_+(S_0/K)) - N(d_+(S_0/U))\big] - Ke^{-rT}\big[N(d_-(S_0/K)) - N(d_-(S_0/U))\big],\\
\mathcal{B} &= U\left(\frac{S_0}{U}\right)^{-2r/\sigma^2} \big[N(d_+(U^2/(KS_0))) - N(d_+(U/S_0))\big]\\
&\quad - Ke^{-rT}\left(\frac{S_0}{U}\right)^{1-2r/\sigma^2} \big[N(d_-(U^2/(KS_0))) - N(d_-(U/S_0))\big],
\end{aligned}$$

with

$$d_\pm(x) = \frac{\ln x + (r \pm \sigma^2/2)\, T}{\sigma\sqrt{T}}. \tag{9}$$

The parameters are $S_0 = 110$, $K = 100$, $r = 0.05$, $\sigma = 0.1$, $T = 2$, $U \in \{160, 170\}$, $n = 2^2, 2^3, \ldots, 2^{10}$, and $M = 10^6$ simulations per case.

Question 2.1. Results and Accuracy

We use $M/2 = 5\times10^5$ antithetic pairs. For each pair j, a single vector of n i.i.d. standard normal shocks $Z^{(j)} = (Z_1, \ldots, Z_n)$ is drawn. Two GBM paths are evolved in parallel using $+Z^{(j)}$ and $-Z^{(j)}$:

$$S_{i+1} = S_i \exp\!\Big[(r - \tfrac{1}{2}\sigma^2)\delta \pm \sigma\sqrt{\delta}\, Z_i\Big].$$

Along each path the running maximum is tracked to check the barrier condition. The payoff is zero if knocked out and $e^{-rT}\max(S_n - K, 0)$ otherwise. The pair average $Y_j = \frac{1}{2}(C^{(j)} + \tilde{C}^{(j)})$ has lower variance than an independent sample of the same size, since $C^{(j)}$ and $\tilde{C}^{(j)}$ are negatively correlated. The estimator and 95% CI are

$$\hat{\mu}_C = \bar{Y}, \qquad \text{SE} = \frac{s_Y}{\sqrt{M/2}}, \qquad \text{95\% CI: } \hat{\mu}_C \pm 1.96\,\text{SE},$$

where s_Y is the sample standard deviation of the $M/2$ pair averages. (See `exercise2_main.m`, `mc_uoc_antithetic.m`, and `fUOC_continuous.m` for the implementation.)

Tables 6 and 7 report the antithetic MC estimates for both barrier levels. Two distinct sources of error are present in each estimate:

1. **Simulation error** (sampling noise), measured by the standard error and the 95% CI width, is controlled by M and the variance reduction method; it does not vanish as n grows.
2. **Discretisation error** (the gap between the discrete and continuous prices), measured in the "Gap to CF" column, is a systematic bias that shrinks as n increases but is unaffected by M.

n	MC Price	Std Error	CI Lower	CI Upper	CI Width	Gap to CF
4	18.6133	0.0063	18.6009	18.6256	0.0247	0.6590
8	18.5267	0.0064	18.5141	18.5394	0.0253	0.5725
16	18.4181	0.0066	18.4051	18.4311	0.0260	0.4639
32	18.3060	0.0068	18.2926	18.3195	0.0269	0.3518
64	18.2098	0.0070	18.1961	18.2236	0.0275	0.2556
128	18.1519	0.0071	18.1380	18.1658	0.0278	0.1977
256	18.0851	0.0072	18.0709	18.0992	0.0282	0.1308
512	18.0604	0.0072	18.0461	18.0746	0.0284	0.1061
1024	18.0279	0.0073	18.0136	18.0423	0.0286	0.0737

Table 6: Antithetic MC estimates for $U = 160$. Continuous benchmark $f_{\text{UOC}} = 17.9542$.

n	MC Price	Std Error	CI Lower	CI Upper	CI Width	Gap to CF
4	19.5144	0.0047	19.5051	19.5237	0.0186	0.2754
8	19.4878	0.0048	19.4784	19.4973	0.0189	0.2489
16	19.4423	0.0049	19.4327	19.4519	0.0192	0.2033
32	19.3937	0.0050	19.3839	19.4035	0.0196	0.1548
64	19.3586	0.0051	19.3486	19.3685	0.0199	0.1196
128	19.3260	0.0051	19.3160	19.3361	0.0201	0.0871
256	19.3133	0.0051	19.3032	19.3233	0.0202	0.0743
512	19.2833	0.0052	19.2731	19.2935	0.0204	0.0443
1024	19.2772	0.0052	19.2670	19.2874	0.0204	0.0382

Table 7: Antithetic MC estimates for $U = 170$. Continuous benchmark $f_{\text{UOC}} = 19.2389$.

For both barriers the CI widths lie between 0.019 and 0.029, giving approximately 2 reliable decimal places throughout. The CI width stays roughly constant as n increases, confirming that increasing n does not reduce simulation error. The discretisation gap dominates total error at small n: for $U = 160$ it is 0.659 at $n = 4$ (roughly 25 times the CI half-width) and falls to 0.074 at $n = 1024$ (still about 5 times the half-width), so simulation error becomes relatively more important as n grows.

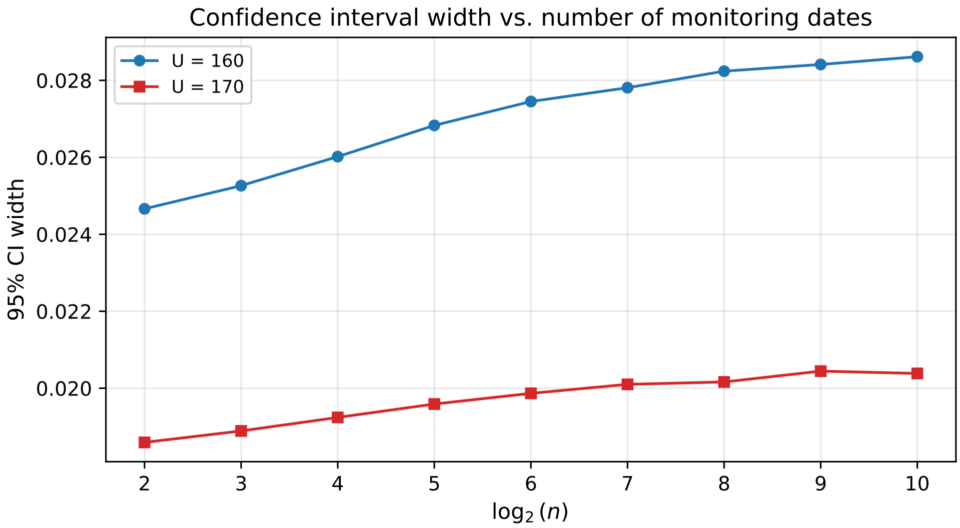

Figure 3: 95% CI width as a function of $\log_2(n)$. The width is nearly flat across all n, confirming that simulation error is controlled by M and the variance reduction method rather than by the monitoring frequency.

Performance of antithetic variates

To quantify the benefit of antithetic variates we also run a crude Monte Carlo estimator at the same total budget of $M = 10^6$ independent paths for representative values of n. Table 8 reports the variance ratio $\mathrm{VR} = \hat{\sigma}^2_{\mathrm{crude}}/\hat{\sigma}^2_{\mathrm{AV}}$; a ratio above 1 means antithetic variates are more efficient.

U	n	Crude SE	AV SE	Crude CI Width	AV CI Width	Var. Ratio
160	4	0.0136	0.0063	0.0532	0.0247	4.65
160	64	0.0134	0.0070	0.0523	0.0275	3.64
170	4	0.0142	0.0047	0.0556	0.0186	8.94
170	64	0.0141	0.0051	0.0551	0.0199	7.70

Table 8: Crude MC vs. antithetic variates at the same total budget $M = 10^6$.

Variance ratios range from 3.6 to 8.9, meaning antithetic variates reduce variance by a factor of 4 to 9. The reduction is larger for $U = 170$: when the barrier is further away, fewer paths are knocked out, the payoff is smoother, and the antithetic pairing exploits monotonicity more effectively. The improvement is less dramatic than for plain vanilla options as the barrier introduces a discontinuity in the payoff; a path just below U receives a positive payoff while one just above receives zero, weakening the negative correlation between antithetic paths.

Question 2.2. Convergence to the Continuous Benchmark

As n increases the monitoring grid becomes finer, more barrier crossings are detected, and the discrete price decreases monotonically towards the continuous benchmark. Discrete monitoring misses crossings between dates, so fewer paths are knocked out and the discrete price systematically overshoots the continuous one. Refining n closes this gap.

For $U = 160$, the gap falls from 0.659 at $n = 4$ to 0.074 at $n = 1024$, a factor of roughly 9 over 8 doublings of n. Examining successive doublings, the gap halves roughly every quadrupling of n, for example from 0.659 ($n = 4$) to 0.352 ($n = 32$) to 0.198 ($n = 128$). This is more broadly consistent with the known $O(n^{-1/2})$ convergence rate for discretely monitored barrier options.

For $U = 170$ the absolute gaps are smaller at every n (0.275 at $n = 4$ to 0.038 at $n = 1024$), since a barrier further from $S_0 = 110$ is crossed less frequently and the discrete and continuous regimes differ less in the first place.

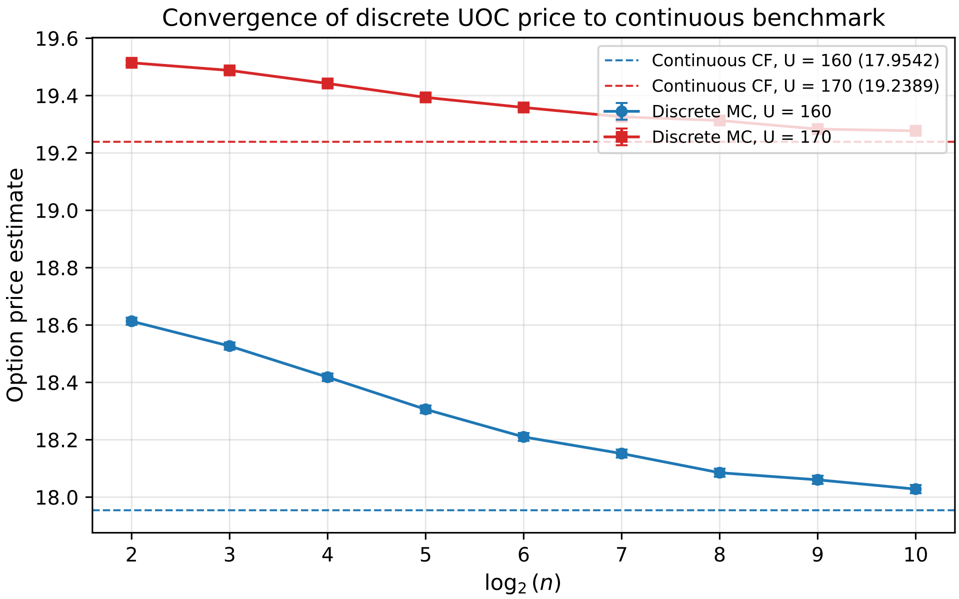

Figure 4: MC price estimates (with 95% CIs) vs. $\log_2(n)$ for $U = 160$ (blue) and $U = 170$ (red). Dashed lines mark the continuous benchmarks. Both curves decrease monotonically from above.

The gap doesn't vanish at $n = 1024$: residual values of 0.074 ($U = 160$) and 0.038 ($U = 170$) remain, and convergence visibly slows at the upper end of the range. This is expected under $O(n^{-1/2})$ convergence, where each additional doubling of n produces a diminishing absolute reduction. The $U = 170$ option is consistently more expensive than the $U = 160$ option because a higher barrier reduces the knock-out probability, allowing more paths to survive to maturity.

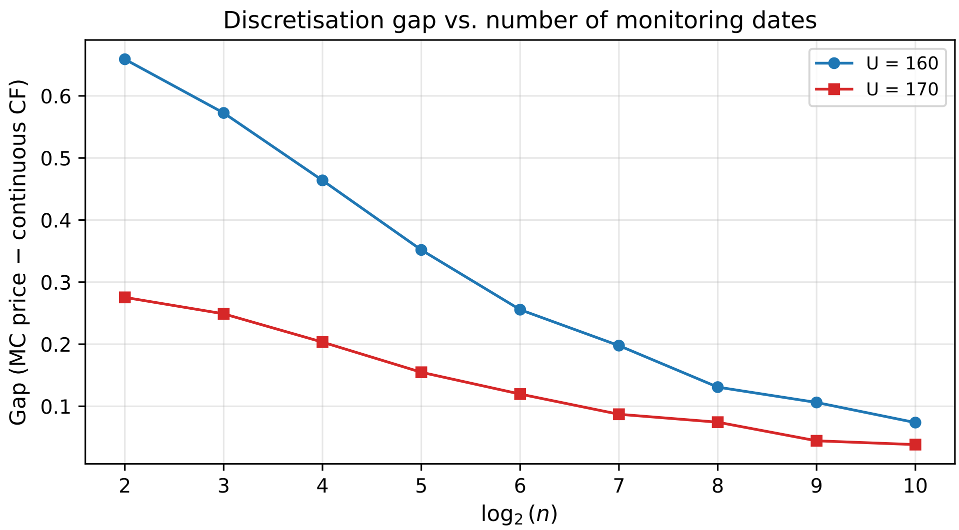

Figure 5: Discretisation gap (MC price − continuous CF) vs. $\log_2(n)$. Both barriers show monotone convergence. The $U = 160$ gap is larger at every n as the closer barrier is more sensitive to monitoring frequency.

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References

Curran, M. (1994). Valuing Asian and portfolio options by conditioning on the geometric mean price. *Management Science*, 40, 1705–1711.

Kemna, A.G.Z. and Vorst, A.C.F. (1990). A pricing method for options based on average asset values. *Journal of Banking & Finance*, 14, 113–129.

Shreve, S. (2004). *Stochastic Calculus for Finance II: Continuous-Time Models.* Springer Finance, Springer.

Thompson, G.W.P. (1999). Fast narrow bounds on the value of Asian options. Working paper, Centre for Financial Research, Judge Institute of Management, Cambridge University.